



GOVERNMENT OF ODISHA
OFFICE OF THE CHIEF ENGINEER-CUM-CHIEF ELECTRICAL INSPECTOR
NORTH ZONE, ODISHA, BALASORE
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ANNEXURE-I
INSPECTION OBSERVATIONS AND RECOMMENDATIONS

1	Report No. Online Application ID:	108/2025-26 INELEIN/2026/00333
2	Date of Inspection	07.03.2026, 16.03.2026, 17.03.2026 (Besides other inspection visits)
3	Inspection Fees	Inspection Fees Due: Rs. 1,62,16,100 /- (provisional & subject to revision) Paid Amount: Rs. 1,62,16,100 /- Reference ID: CK00OAMTE9, Payment Date: 26.02.2026
4	Type of Inspection	ANNUAL
5	Voltage and System of Supply	220 KV, 33 KV, 11 KV, 6.6 KV, 415 V, 3 Phase AC
6	Name of the Consumer & Address	M/s Jindal Stainless Limited, KNIC, Danagadi, Dist- Jajpur, Odisha- 755026.
7	Location of the Installation	Inside the Factory Premises
8	Particulars of the Installations:	ANNEXURE-B
9	Inspection Conducted by:	Sri A. Behara, CE-cum-CEI, North Zone, Balasore
	Assisted by	Sri B. K. Sahoo, TA to CE-cum-CEI, North Zone, Balasore Sri P. K. Jena, AEE, O/o CE-cum-CEI, North Zone, Balasore Sri Birendra Kumar Rout, Electrician Gr-II
10	Persons Present	Sri Pradipta Kumar Das, DGM, Power Plant Sri Prabhat Kumar Sahoo, Sr. Manager, MRSS Sri Manas Kumar Jena, Manager and other Heads / representatives from different units (during different visits).
11	Single Line Diagram	Updated SLD and plant layout diagram indicating the electrical installations as per actual installations need to be maintained. SLDs need to be displayed in all control rooms. A layout diagram showing individual earth electrodes/grid and main earth connection is to be maintained for each functional area.
12	INSPECTION OBSERVATIONS <i>The electrical installations are still to be maintained as per prescribed Safety Standards and Safety Regulations besides applicable Indian Standards and the manufacturers' guidelines, especially the transformers and associated switchyards, control rooms, cables, OH lines/underground cables inside the plant premises, motors etc. The observations and deficiencies mentioned have already been explained to your representatives present during the inspection visit for compliance. Specific attention is invited regarding fire safety of transformers as per regulation 46, CEA (Measures Relating to Safety and Electric Supply) Regulations 2023. Attention is also invited to Regulation 48(7) regarding the responsibility of owners. The last inspection observations have been reiterated for ready reference since compliance to some of the observations are to be completed and are also generic in nature for periodic review.</i> • The HVDB-1 room (COBP Area) shall be provided with an air conditioned/ventilated system to keep the room dust free. • New DLMS compliant Energy Meter/CT/PT of class 0.2 s shall be installed for the 6 MW DG set in place of the old energy metering system and compliance to be submitted. • <i>OTI & WTI viewing glass of the marshaling box of many transformers should be replaced immediately.</i> • <i>Pending NIPFS/HPWS works for some of the power transformers inside MRSS needs to be completed on priority. The fire wall height is to be increased as per the transformer height.</i> • <i>All the Power transformers at MRSS are to be cleaned (Cleaning and painting if required). This will help in identifying in case of oil leakage in future.</i> • The energy metering system (Energy Meter, CTs, PTs) of the TG1 and TG2 are to be replaced with 0.2s class DLMS compliant energy metering system on priority.	



- *Broken and missing slabs in the switchyard's cable trench (Inside and outside) are to be replaced.*
- *Transformer cleaning to be done & also oil leakage to be arrested.*
- ***Dust deposits in Transformers and allied installations needs to be removed at regular intervals.***
- *Lubricant oil storage is to be handled safely and the oil spillage near the DG area is to be avoided.*
- ***Specific attention is invited for regular maintenance of Furnace as per manufacturer guideline and compliance is to be submitted to this office.***
- *Cable cellars are to be safely maintained with adequate fire protection measures.*
- ***It is observed that monthly ED returns for some of the new DG sets are not being communicated. It needs to be communicated. by JSL.***
- *Oil leakage in MRSS and Power Plant transformers to still be addressed. The pits below the transformers are to be cleaned and drainage arrangements to be ensured. Areas to be cleaned of grass and weeds. Cable termination of LV sides are to be sealed, tagged and identified as already communicated earlier.*
- *Transformers (ground cover) made with concrete are impervious and will support spill fire. Impervious surfaces can allow the burning oil to form a large pool fire, which will increase the heat flux to adjacent equipment and structures. The use of 30 cm (12 in) thick crushed stone ground covers below the transformer will suppress the flames from a burning oil spill fire. Refer to IEEE Std 979™-2012: IEEE Guide for Substation Fire Protection.*
- *Oil leakage from the 120 MW GT and other transformers is to be checked on priority. The area below the transformers are soaked with oil and need to be attended to avoid fire risk.*
- *The name of the Electrical Safety Officer and designated persons as per regulation CEA (Measures Relating to Safety and Electric Supply) Regulations 2023 is to be communicated to this office. Please refer to Regulation 3 to 6 of the Regulations in this regard and ensure compliance.*
- *The oil leakage from the GT, UAT and other transformers are to be attended and the soaked gravel and soil below the transformers are to be replaced to reduce fire risk.*
- ***The prescription as made in Regulation 46(ix) of CEA (Measures Relating to Safety and Electric Supply) Regulations 2023 regarding provision of fire detection, alarm and protection system, fighting system as per IS-3034:1993 or with Nitrogen Injection Fire Protection System has not been implemented for many of the transformers in the plant of capacity 10 MVA or higher or with oil capacity of more than 2000 ltrs. It is highly recommended to install firefighting systems as prescribed in the Safety Regulation and applicable Indian Standard.***
- *The 16 MVA, 33/6.6kV S/stns of the SMS area are to be maintained with working fire protection systems, proper access. The earth electrodes are to be protected from damage.*
- *Since oil leakage of the GT/UAT has been continuing, it is advised to remove the oil soaked ground cover and replace with clean stone to prevent spread of fire. The oil leakages are yet to be addressed.*
- *Cable entries into the power plant building are to be sealed as per fire safety measures mentioned herewith.*
- *The rusted NGRs are to be painted and labeled.*
- *The entire area of the substation/ switchyard shall be maintained with 150/100mm thick gravels to restrict the growth of grass in the outdoor substation/ switchyard. The gravels shall consist of 75/50mm thick of 40mm stone size on the top and 75/150mm thick of 20mm stone size below. The resistivity of gravel shall be 3000Ω-mtr. The area shall be cleaned of grass and weeds.*
- *The Pump house for SAF-1 & 2 primary pumps, transformer pumps are to be safely maintained, panels to be identified, cables are to be safely laid and protected. Caution to be exercised to avoid using temporary connections without adequate safety and protection from electric shock.*
- *The electrical installation of the Metal Recovery panels is to be safely maintained.*
- *Applicable safety measures need to be ensured for all the Furnaces for protection from fire hazard and gas as per prescribed statutory provision. Safety measures for control rooms and work areas are to be ensured. The HVWS fire protection was found inoperative and need to be attended on priority. The Transformers are to be identified.*
- *The 12 MVA 33/6.6 kV (S/stn-3) besides the access road of JSW Cement is found to be unsafe and need to be properly maintained as per the guidelines and observations mentioned herewith. The Transformers, Cables and Switchyard area is to be addressed on priority.*
- *Interlock needs to be ensured for the Star-delta change over switches of the SAF.*
- *Safe electrical clearance is to be ensured for the 160 MVAR SVC of the SMS.*
- *Implementation of QR code-based access to drawings and other technical parameters on panels and equipment can be explored for providing instant, online and easy access to drawings and required technical details.*



	- Transformers in the HSM area need to be maintained as per prescribed Indian standard. <i>Latest Periodical testing certificate and condition monitoring test report of the transformers to be submitted on priority.</i> - Attention is invited to the National Electrical Code of India 2023, SP 30: 2023, which has been released in January 2023. JSL authorities are advised to adopt the Standards and recommendations prescribed in the Code with consideration for safety and economic usage of energy in the design, execution, inspection and maintenance of electrical installations of various locations. Since JSL is undertaking Solar Installations, Part-8 of the NEC shall be useful. Also, attention is invited to lightning protection, temporary installations, installations in hazardous areas etc. - Specific attention is invited to the electrical installations of the Ferro Alloy installation, SMS areas, RMHS etc. such as the Distribution Transformers, Furnace Transformers, Panels in the furnace area, etc. The installations are to be safely maintained as per prescribed standards. The HVWS pipe lines are to be colored RED as per IS.12349, 33kv identifications to be done for different SAF transformers etc. The S/stn areas of the transformers are to be maintained with appropriate fencing access and proper laying of cables etc. Many of the transformers are covered with excessive dust and need to be cleaned periodically. The granite layer within the transformer S/stns are completely buried inside dust and need to be addressed. Access to many auxiliary transformers such as melting shop cranes need to be ensured. Bus duct entries into sub stations, fire protection of cables in the cable cellars is to be addressed. - It is also to inform that for lightning protection, National Building Code of India 2016, PART 8 BUILDING SERVICES — SECTION 2 ELECTRICAL AND ALLIED INSTALLATIONS 11.5.1.1 Any other kind of air-terminal like dissipation system/ESE air-terminal/CSE air-terminal shall not be acceptable. Hence their use for solar installations may not be accepted. - The dry type indoor transformers are to be adequately labeled with voltage and capacity also for HV and LV. The side doors are to be suitably locked and indicated that live HV terminals are inside and the door interlock operations for tripping to be ensured. The cables from these transformers are to be tagged and identified. - The water drainage arrangement from the cable cellars are to be ensured. - The auxiliary transformers of the SAF are to be properly maintained and the accumulated dust is to be periodically cleaned. Oil leakages to be addressed. - The HVWS pipe lines sometimes hinder access to transformers and need to be addressed. Clear passage around the transformer needs to be ensured for periodical inspection. - Periodical testing of the fire protection systems, specifically for the cable cellars of SAF control room and other locations need to be undertaken. - For air-conditioning systems circulating air to more than one floor area, it should be provided with dampers designed to close automatically in case of fire and thereby prevent spread of fire or smoke. Such a system should also be provided with automatic controls to stop fans in case of fire, unless arranged to remove smoke from a fire, in which case these should be designed to remain in operation. Please refer to IS Clause 8 of 1642.1989. - All openings for cable entries/exits can be provided with fire barriers to prevent spread of fire. Use of Linear heat sensing devices can be used for the cable running from the TGs to the control panels. Recommendation of IS 12459 (1988): Code of Practice for Fire Safety in Cable Runs need to be adopted for cable systems.
A	220KV MRSS, Switchyard - All open cable trenches, water drainage pit and other openings to be covered and protected. - Storage of inflammable materials inside the MRSS building and near the cable vaults such as paints are to be avoided. Such stores are to be relocated. Earthing of all cable trays and support structures are to be ensured. - Cables shall be laid, maintained and protected as per IEC 62067 and other applicable IEC standards. The cable trays supporting the HT cables in the switchyard are to be provided with continuous earthing. Control and other power cables are to be laid and maintained as per IS 1255. - Availability of adequate safe working and ground clearance needs to be ensured at the switchyards and associated lines. Please refer to regulation 46 and Schedule- V of Regulation CEA (Measures Relating to Safety and Electric Supply) Regulations 2023. - The outdoor panels/marshaling box connectivity to earth to be maintained. The panel doors are to be connected to earth using flexible copper bonds. Interlock and protection arrangements to be ensured as per Regulation 47 of CEA (Measures Relating to Safety and Electric Supply) Regulations 2023. - The damaged insulating mats provided in the control rooms are to be checked and replaced if necessary.



- Protecting shed to be provided for the sand buckets and fire extinguishers near the transformers/switchyard to be maintained. Adequate no. and type of fire extinguishers to be provided.
- The 500 KVA 33/0.433 kV Station Transformers are to be maintained as per IS 2026. The fencing is to be connected to earth. The outgoing cables are to be safely laid and protected. The HV and LV cable boxes are to be provided with voltage level. The cables are to be tagged and identified.
- HT cables shall be identified with cable tag at both the ends as well as throughout the cable routes. Cable route to be identified with route markers along with danger notices indicating the presence of HT cables. The cable support structures shall be identified with voltage level for HT cables.
- Phase marking and identification of bays to be done for all the bays with proper nomenclature of voltage level etc. The markings have faded over time.
- The outdoor panels to be provided with voltage level and details. The cables are to be tagged and identified and rusted covers are to be painted.
- The interlock arrangements to be periodically verified. Specific attention is invited to the provisions in regulation 47 of CEA (Measures Relating to Safety and Electric Supply) Regulations 2023 regarding isolators, breakers and earth switches. Both electrical and mechanical interlock needs to be ensured. LOTO systems to be adopted for safety.
- The bays, structures, equipment such as isolators, breakers, earth switch etc need to be properly marked in the Switchyard for ease of identification. Bays are also to be clearly identified.
- The Switchyard lighting power supply cabling arrangements to be safely done. Joints and loose connections to be attended. Cables to be properly routed and supported.
- The temporary power supply sockets provided in the marshaling box are to be mandatorily provided with RCCB for safety.
- As per Indian Standard 15652 (2006): “Insulating mats for electrical purposes” needs to be provided.
- The CEA (Technical Standards for connectivity to the Grid) Regulations 2007, specifically PART-III pertaining to protections, substation auxiliary supply needs to be ensured.
- IS 12459 (1988): Code of Practice for Fire Safety in Cable Runs need to be considered for cables. Use of LHS and fire-retardant paints, fire barriers are to be considered.
- Safe working clearance in the Switchyard is to be ensured. Areas with lower clearances are to be identified and clearly marked till rectification
- Cable entry or exit into the marshalling/console box to be properly done using cable glands and the unused openings to be properly sealed. IS 1255 needs to be adopted for installation and maintenance of cables. The control cables are to be properly dressed, tagged and routed and cable support trays to be provided as necessary. Adequate clearance needs to be ensured between the live phase connections and inside the panels.
- Use of Linear heat sensing devices can be used for the cables inside the cable gallery. HT cables are to be identified.
- The entire area of the switchyard is to be cleaned of vegetation and the metal cover of granite metal to be restored to ensure proper covering for safety of manpower.
- It needs to be ensured that the earthing of marshaling boxes, control panels and equipment are always made with two separate earth connections from distinct sources. Please refer to clause 4.2 of IS 3043 regarding the statutory provisions for earthing. Some of the marshalling box/ control panels (e.g. breaker) are found with single earth connections.
- The control panels for the lighting arrangement of the switchyard needs to be properly maintained and connected to earth.
- The Lightning Arresters need to be verified for proper operation. The leakage currents to be checked to be within the prescribed range. Refer to the Regulation 46(8)(e) of Central Electricity Authority (Technical Standards for construction, operation and maintenance of electrical plants and electric lines) Regulations 2022, regarding provision of surge counters and arresters. The reading of surge counters can be recorded periodically to ascertain proper functioning of the surge arresters.
- All earth electrodes are to be tested and verified. Test results are to be submitted. Damaged dysfunctional earth electrodes are to be replaced.
- The earth pits are to be numbered and indicated with individual and combined resistance values and date of last testing. Records of the details need to be maintained. An updated drawing showing the main earth connection and electrodes to be maintained as per clause 4.7 of IS 3043:2018.
- Cables used for extending temporary power supply to be laid and supported as per IS 1255. Cables to be tagged and identified.



	• Auxiliary earth mats are to be maintained to cover the area below the isolator handle for better safety. The operating handles are to be connected to earth using flexible copper bond. Appropriate touch and step potential need to be ensured since the earth mat design has not been submitted. • The entire area of the substation/ switchyard shall be maintained with 150/100mm thick gravels to restrict the growth of grass in the outdoor substation/ switchyard. The gravels shall consist of 75/50mm thick of 40mm stone size on the top and 75/150mm thick of 20mm stone size below. The resistivity of gravel shall be 3000Ω-mtr. • Adequate fire safety of control rooms is to be ensured. You may refer to IEEE Guide for Substation Fire Protection and IEEE Guide for Containment and Control of Oil Spills in Substations besides other standards.
B	80 MVA, 120 MVA 220/34.5 kV Power Transformers
	<i>The physical distance of the transformers from other transformers installed in the Grid S/stn needs to be clearly indicated and accordingly provisions of fire wall and fire safety is to be made if physical distance is less than 15m. Please refer to the Central Electricity Authority (Technical Standards for construction of electrical plants and electric lines) Regulations 2022. (80 MVA and 120 MVA Trf.)</i> • As per sub regulation 8 of Regulation (48), CEA (Measures Relating to Safety and Electric Supply) Regulations 2023 format within one month of the occurrence of failure, and need to be strictly complied. • The transformer enclosure fencing gates to be connected to earth using flexible copper bonds. Fencing earth needs to be periodically tested for continuity. • Appropriate fire safety measures to be taken for transformers having more than 2000 ltrs of oil as per CEA (Measures Relating to Safety and Electric Supply) Regulations 2023. • The outgoing cables are to be tagged and identified. Fire retardant paints and LHs can be considered for safety. The rusted cable support structures are to be painted and earthed. Voltage levels are to be indicated. The switchyard area near the transformer bays are to be properly maintained. The granite materials are found buried within soil and shall not provide the required protection and may increase the step potential. • The high velocity water pipes provided for fire protection of transformers (e.g 80 and 120 MVA) are to be repainted as per IS. Spilled oil below the transformers (120 MVA) are to be cleaned to reduce fire hazard. • Silica gels of transformers are to be checked and replaced. • The adequacy of the water spray system to be verified and tested as per IS 3034. • Since there is a large volume of oil in the transformers, oil drainage arrangement needs to be verified. The capacity of the oil soak pit should be more than the total oil volume of the transformers. Adequate capacity and size of transformer oil soak pit needs to be ensured as per CEA Safety Regulation 46 and proper size of drainage arrangement is to be made as per IS 3034. • The steel ladders provided for the transformers are to be connected to earth and kept locked. • The control cables of the transformers are to be supported, dressed and supported. • The drainage passage from the transformers to the drainage pits are to be marked in the switchyard layout diagram and identified. Oil soak pits are to be clearly identified with capacity and drainage pumps are to be provided for the oil soak pits. Oil drainage pipes and pits are to be cleared of chokes to facilitate oil drainage. • Oil leakages from the transformers e.g. top cover, bushings etc. are to be rectified. The transformers are to be kept clean to facilitate identification of oil leakage. • The Lightning Arresters provided for the transformers need to be verified for proper operation. The leakage currents to be checked to be within the prescribed range. • Unused equipment is to be stored and maintained as per manufacturers' guidelines. The unused transformer bay is to be cleaned of trees and vegetation and maintained. • Guidelines as recommended by the respective manufacturers of the equipment and statutory safety practices need to be adopted for the operation, maintenance and periodic testing of transformers. • The protection and interlock arrangement of the transformers also to be periodically verified for correct operation. • The silica gel to be periodically checked and replaced. • It needs to be ensured that the earthing of marshalling boxes, control panels and equipment are always made with two separate earth connections from distinct sources. Please refer to clause 4.2 of IS 3043 regarding the statutory provisions for earthing.
C	E.S.P. Installations
	• Danger Boards are to be provided at the E.S.P. Transformer. Control and power cables to be laid separately for the ESP transformer. Cable tag markers at regular intervals are to be provided both for control & power cables.



	- Lightning spikes at suitable intervals on the metal structure over the E.S.P. transformers can be provided as necessary. - I.D. fan motors are to be double earthed and their earth connections to be verified and properly routed. - Operation of Mechanical and Electrical interlocking and control systems is to be ensured. - Cable trays to be provided for carrying power and control cables to the transformers and are to be earthed by using adequate size of earth flat. - The ESP floor cables to be routed or supported. The silica gel of the transformers to be changed and oil level to be maintained. - Earth flats are to be laid properly, joints if any to be welded. - The ESP transformer units are to be maintained as per prescribed guidelines. The ESP being a critical installation for environmental and pollution compliance, needs to be maintained regularly for required operation. - The MV control/main switches on the ESP floor are to be properly maintained and earthed. The connections are to be properly done using appropriate connectors rather than being loosely twisted. Entry/exit of cables from the panels/switches to be done using cable glands, gland covers and properly sealed. MCCBs to be properly mounted.
D	Furnaces, transformers and control panels
	- Adequate safety measures need to be taken in consultation with the manufacturers to prevent fire due to leakage of oil from the furnace transformers since the volume of oil is more and the transformers are housed inside closed areas. Adequate fire alarm, detection and ventilation arrangements need to be made and their operations need to be ensured. - Provision of oil drains and soak pits needs to be considered as per prescribed standards. - Working space around the control panels and control areas needs to be ensured as prescribed under CEA safety regulation 39. Storage of materials needs to be avoided for the purpose. - The furnace components should meet safety requirements specified in IS: 9080 (Part 1)- 1979, IS: 9080 (Part 4)- 1981 and other applicable standards. - Storage of materials needs to be avoided in the furnace transformer area. The illumination level in the furnace transformer area needs to be improved. - Appropriate fire safety measures to be taken for transformers having more than 2000 ltrs of oil as per CEA (Measures Relating to Safety and Electric Supply) Regulations 2023.
E	11/6.9kV 20 MVA UAT, 11/235 kV 150 MVA GT, 33/6.9kV 30 MVA ST (125 MW PP) and 1.6 MVA UATs, 16 MVA GT (13MW PP), Auxiliary Distribution Transformers
	- Oil leakage from the GT near the radiators and oil pumps is to be addressed. - The defective viewing glasses of the marshalling box for WTI and OTI are to be replaced. - The drainage passage from the transformers to the drainage pits are to be marked in the switchyard layout diagram and identified. Oil soak pits are to be clearly identified with capacity and drainage pumps are to be provided for the oil soak pits. - Since there is a large volume of oil in the transformers, oil drainage arrangement needs to be verified. The capacity of the oil soak pit should be more than the total oil volume of the transformers. Adequate capacity and size of transformer oil soak pit needs to be ensured as per CEA Safety Regulation and proper size of drainage arrangement is to be made as per IS 3034. - The rusted cable support structures for the GTs are to be painted and earth connection is to be ensured. Rusted transformers are to be painted. The HV side, LV side and marshalling box of the transformers are to be labeled with transformer No and voltage level. - Guidelines as recommended by the respective manufacturers of the equipment and statutory safety practices need to be adopted for the operation, maintenance and periodic testing of transformers. - The protection and interlock arrangement of the transformers also to be periodically verified for correct operation. - The silica gel to be periodically checked and replaced. - The fencing provided for the transformers including the gates to be connected to earth and verified for connectivity. - The outgoing cable from the transformers to be tagged and identified, cleaned of oil spillage. Fire retardant paints can be applied for safety. Outgoing cables to be identified and painted with appropriate color for indicating the phases. - The steel ladders provided for the transformers are to be connected to earth and kept locked.



	• The control cables of the transformers are to be supported, dressed and routed. • The entire switchyard is to be cleaned and granites are to be uniformly spread as mentioned before. All rusted structures are to be painted. Phase and voltage labeling is to be carried out. • The 33kV cable routes are to be clearly identified and protected. • Oil leakages from the transformers are to be checked (GT, ST, etc). The transformers to be kept clean for ease of identifying oil leakage. (e.g. from top covers, flanges, LV Box etc.) Transformers are not safe to be operated with the existing oil leakages and to be addressed on priority. Oil level to be maintained for all the transformers. • The unsafe opening near the transformers is to be covered. The rusted cable support structures are to be painted and earthed. • The physical distance of the transformers from other transformers installed in the Grid S/stn needs to be clearly indicated. Please refer to the Central Electricity Authority (Technical Standards for construction of electrical plants and electric lines) Regulations 2022. Firewalls are to be indicated in the layout plan. • The windows/doors and other openings between the TG building and the transformers are to be closed or provided with adequate fire rating doors. • The 1.6MVA UATs WTI and OTIs to be checked are replaced. The LV box cable terminations are to be sealed, tagged and identified. The cable tray earthing is to be ensured. • The 1.6 MVA SAT is in bad shape and is to be safely maintained. The cables near the HV bus duct are to be properly laid in trays. Power and control cables are to be separated. • The 16 MVA GT of the 13 MW PP is to be safely maintained. The S/stn granite is to be maintained, the damaged fencing is to be repaired and earthed, High velocity water protection system is to be provided, the cables are to be laid and protected properly using cable trays or inside trenches etc. The earth electrodes and pits of the transformer are to be maintained. Silica gels to be periodically checked and replaced. The cable terminations of the transformers are to be sealed. Oil leakage from radiators to be checked. Control and power cables are to be dressed and protected. • The earth electrodes of the transformers are to be maintained. Rusted connections are to be attended. • The LA and associated surge counters are to be checked for proper operation or replaced.
F	Power Plants, 2 x 125 MW TG, 13 MW TG and Associated Auxiliaries • The TG nameplate rating indicates 120 MW which differs from the label provided on the TG sets and needs to be clarified. • Proper IP rating to be ensured for the terminal cover of the condensate extraction pumps. • All cables to be properly tagged with cable tag at both the ends as well as at intermittent points. HT cables and their routes to be specifically marked at specific intervals. • Oil leakage and vibration of the BFPs to be checked. Vibration to be maintained within prescribed limits. • The earth connection of the TGs at the designated points to be verified. • Adequate number of fire escape staircases shall be identified and marked in the TG building. • Generator terminal covers to be indicated with voltage level and phase indication of cables. • Linear Heat Detection (LHD) devices (also known as Linear Detection Wire or Linear Heat Detection Cable) can be used for the cables used in the power plant control rooms which have a higher density of cables. Cables need to be laid and maintained as per IS 1255. • The earth pits are to be checked for proper functioning. Many of the earth electrodes are found to be deeply buried and not easily accessible for inspection. Their mechanical strength and conditions needs to be checked and replaced if necessary. Please refer to IS 3043 for the purpose. Availability of earth test links to be considered as per section 4.6 of IS 3043:2018. • Water drainage arrangement near the condensate pumps to be ensured. • All the motors, compressors, machines, metallic parts etc. not intended as conductors, iron support structures etc. shall be earthed by two separate and distinct connections with earth. Frame of every metallic part not intended as conductors, iron support structures shall be earthed by two separate and distinct connections with earth. • The power plant steel structure is to be effectively used as an equipotential bonding and properly connected to earth grid. The provisions of IS 3043 are to be adopted. • Cable entry or exit into the control rooms, panels, motor terminals, marshalling/console box to be done using cable glands and the unused openings to be properly sealed. IS 1255 needs to be adopted for installation and maintenance of cables. • The TG floor openings near the pipe entry/exit, equipment to be properly covered.



- The Steam inlets and outlet pipes insulations are to be checked and verified.
- The TG unsafe areas to be barricaded to prevent unauthorized access.
- All parts of the steam generation system including pressure parts, vessels, piping, valves including safety valves and fittings shall meet the requirements of Indian Boiler Regulations.
- The turbine protective devices such as the over speed trip device and emergency hand trip device both locally and from the control room need to be verified for proper operation.
- The LAPT and NGR panels are to be properly identified along with voltage level and their earthing is to be ensured.
- Statutory compliance to the provisions as prescribed in the Central Electricity Authority (Technical Standards for Construction of Electric Plants and Electric Lines) Regulations 2022, besides other applicable standards needs to be ensured for the power station.
- All unsafe openings, open trenches etc. within the power house, cable galleries etc. to be closed/barricaded.
- The emergency stop switch is to be clearly highlighted and can be relocated to an accessible area. Also, the emergency stop control needs to be provided in the control room and properly highlighted with fluorescent tape/color.
- The cable support structures should be provided with continuous running earth. Power and control cable should be segregated and clearances of power cable from control and communication cables to be maintained as per IS 1255.
- Working space around the control panels and control areas needs to be ensured as prescribed under CEA safety regulation 39. Storage of materials needs to be avoided for the purpose.
- The false ceiling of the control rooms needs to be periodically checked. False ceilings are to be provided with inspection openings and the wiring/connections inside false ceilings shall be regularly inspected for safety.
- Adequate illumination is felt necessary for the power house working areas. Sufficient number of lights with backup supply for using during power failure or emergency needs to be ensured. Please refer to the National Building Code for guidelines regarding illumination levels. Adequate number of illuminated exits signage with backup supply can be provided for the Power House and other areas.
- Thermo graphic cameras to be used for mapping of heat level and heat leakage from pipes, for monitoring of bearing/coupling temperatures.
- Single Line Diagram showing all the electrical installations clearly indicating isolating arrangements to be displayed in the control room.
- All HT cables to be properly tagged and identified. Power and control cables are to be segregated.
- Fire detection systems need to be periodically tested. Use of Linear heat sensing devices can be used for the cable running from the TGs to the control panels.
- Adequate fire safety measures need to be ensured for safety of persons working in the power plant areas and the associated control rooms. Fire exits, illuminated exit signs, adequate number of fire extinguishers, fire detection and alarm systems need to be ensured as per statutory provisions. Please refer to the PART-A, general technical requirements regarding the provision of fire escape staircases, elevators etc of the CEA (Technical Standards for construction of electrical plants and electric lines) Regulations 2022. The provision regarding fire detection, alarm and protection systems needs to be maintained in functional condition. Please refer to IS 3034 and applicable standards. Please refer to section 10.2.1.1 regarding requirements of fire fighting appliances and equipment.
- Two or more gas masks need to be provided for each of the power plants as per the Safety Regulation 29(4).
- The safety provisions need to be in conformity with the provisions laid down in the National Building Code and other applicable codes.
- Adequate number of fire extinguishers needs to be placed at different accessible locations in the TG building, specifically near the oil tanks and other lube oil storage areas.
- The BFPs and JB/Panels for the BFPs to be suitably marked with voltage level. Cable entries to the BFP HV terminals are to be sealed. Cables are to be supported and tagged. Direction of rotation to be repainted.
- The emergency push buttons provided for the BFPs and other installations are to be kept at accessible locations and are to be relocated as necessary.
- Unused and spare cables stored in the BFP area are to be removed.
- The electrical systems of the hoists and cranes are to be maintained as per IS 3177, IS 4137 and other prescribed standards. Statutory safety inspection needs to be undertaken for the hoists and cranes as applicable. Periodic testing of hoists/ EOT cranes etc. needs to be ensured as per standard.



	Air Conditioners of the control room shall be invariably provided with overload protections so as to ensure tripping during overloads and faults. The load-parameters mainly load current are to be monitored to ensure proper operating parameters as per manufacturer's guidelines
G	HAPL, CAPL, CRM, HSM, COKE and allied installations
	- The fencing along with the metal enclosures are to be connected to earth. - Adequate fire safety measures are to be ensured for the HAPL control rooms. Adequate number of fire exits are to be ensured. Fire exits are to be clearly illuminated with emergency backup supply. - The cable entry to the control panels of the mill area is to be safely made and the support trays are to be earthed. - All temporary power supply arrangements are to be made through RCCB and using industrial sockets. - High velocity water pressure system is to be implemented for the transformers as already mentioned before, 25 MVA RM/FM transformers, 12.5 MVA WCS & Down Coiler etc. - The earth electrodes are to be verified and replaced if necessary. - All transformers are to be maintained as per IS 2026. - Oil leakage from the converter transformers is to be rectified e.g. 4.2 MVA(HAPL), CAPL 5.8 MVA etc. The granite provided for the S/stns is to be dressed. All granites are found buried in dust. The areas around the transformers are to be cleared of unwanted items. - Fire protection of cable runs are to be ensured in the HT cable vaults.
H	Ferro Chrome Installations, Aux Transformers
	- The S/stns for extending power supply needs to be maintained IS 2026 and manufacturer's guidelines. The fencing earth connection is to be ensured and gates to be kept locked to prevent unauthorized access. - Fencing for the capacitor bank switchyards to be provided and the safe working clearance is to be maintained. Voltage level and danger notices are to be provided. Rusted structures are to be painted. - The power cables are to be placed in cable trenches or using cable trays. Please refer to IS 1255 regarding installation guidelines for cables. - All cables are to be properly tagged and identified. Unused cables are to be dismantled and maintained safely. (e.g. CSS -05) - Cable support structures are to be earthed. Cable tags are to be provided clearly indicating the presence of HT cables as applicable. - Earth pits and electrodes inside the S/stn areas are to be maintained for required earth value. - Unused openings of the panels/kiosks to be closed. Earthlings of all panels are to be made with two separate earth connections from distinct sources. - Power supplies for construction works through PDBs are to be safely maintained and cable joints to be protected. Cables to be safely laid inside trenches or using trays. RCCBs are to be mandatorily used. Portable equipment's earthing needs to be ensured. - Metal sheds provided for temporary installations are to be connected to earth. All metal/pipe fencing provided for electrical installations are also to be connected to earthing.
I	SMS Furnace Transformers and allied installations
	- The fencing provided for the transformer LV bus is to be maintained and connected to earth. Danger notices and voltage level are to be provided for safety. - Oil leakage from the furnace transformers are to be checked. The unused surge diverters are to be removed. - Adequate safety measures need to be taken in consultation with the manufacturers to prevent fire due to leakage of oil from the furnace transformers since the volume of oil is more and the transformers are housed inside closed rooms. Adequate fire alarm, detection and ventilation arrangement needs to be made and its operation needs to be ensured. - Provision of oil drains and soak pits needs to be considered as per prescribed standards. - Working space around the control panels and control areas needs to be ensured as prescribed under CEA safety regulation 39. Storage of materials needs to be avoided for the purpose. - The furnace components should meet safety requirements specified in IS:9080 (Part 1) - 1979, IS: 9080 (Part 4) - 1981 and other applicable standards. - Storage of materials needs to be avoided in the furnace transformer room. The illumination level in the furnace transformer areas, control rooms and panels need to be improved.
J	RMHS and other installations
	Working space around the control panels and control areas needs to be ensured as prescribed under CEA safety regulation 39. Storage of materials needs to be avoided for the purpose.



	• Cables running adjacent to the kilns and hot areas are to be protected from heat. • Cable cellar Room- Dewatering shall be done at DRI Plant • DG if not used – NOT IN SERVICE to be mentioned • Cleaning of transformer rooms at DRI to be carried out of unwanted materials. • Leakage in transformers to be checked. Sturdy barrier for furnace transformer to be provided. Gas mask & Fire extinguisher to be kept at complex-2 • Fencing earthing is to be checked at Coke Oven. • Proper functioning of interlocking and protection arrangements of DRI transformers and other equipment need to be ensured. • All the auxiliary Transformers are to be maintained as per standard and checked for oil leakage. • For transformers with more than 2000 liters of oil, adequate safety measures for fire safety needs to be ensured as per CEA Safety Regulation, CEA Construction Standard and IS 3034 as already mentioned. • All HV/MV cable inlets/outlets to be properly sealed. All cables to be tagged and transformers to be properly labeled on the HV and MV side with voltage level. • The winding and oil temperature indicators are to be preferably relocated to the front side for safe access, visibility and reading. • All the motors, compressors, machines, metallic parts etc. not intended as conductors, iron support structures etc. shall be earthed by two separate and distinct connections with earth. Frame of every metallic part not intended as conductors, iron support structures shall be earthed by two separate and distinct connections with earth. • Unused openings in control panels/ distribution boards shall be sealed. All panel/board covers shall be connected to earth using flexible copper bonds. • The plant steel structure is to be effectively used as an equipotential bonding and properly connected to earth grid. The provisions of IS 3043 are to be adopted. • Cable entry or exit into the control rooms, panels, motor terminals, marshalling/console box to be properly done using cable glands and the unused openings to be properly sealed. IS 1255 needs to be adopted for installation and maintenance of cables. • Cables running on the back side of the DBs/panels to be properly arranged. All cables to be properly tagged with cable tag at both the ends as well as at intermittent points. HT cables and their routes to be specifically marked at specific intervals. • The cable support structures should be provided with continuous running earth. Power and control cable should be segregated and clearances of power cable from control and communication cables to be maintained as per IS 1255.
K	Control Rooms and Control Panels
	• All the panels are identified with proper voltage. SLD and Shock treatment chart is to be fixed at the conspicuous places. • RCCBs are to be provided for the industrial socket in place of MCCBs available in the control room/panels meant for extending temporary power supply. • All panels shall be provided with the description of its identification at front and at the rear along with voltage level. • Regular checking for sparks, damaged insulations and loose connections etc. are to be taken up for all terminations/joints and attended immediately. Thermo vision scanning can be useful for detecting hot spots. • Indicators and alarm for main supply, control supply, motions, wind pressure, phase failure, etc, shall be provided in the operator's cabin. • SLD diagram to be provided in the control room. Emergency switches to be provided with fluorescent markers for visibility during power failures.
L	HT/MV Motors, Drives and HT/MV Loads
	• All the motors, compressors, machines, metallic parts etc. not intended as conductors, iron support structures etc. shall be earthed by two separate and distinct connections with earth. Frame of every metallic part not intended as conductors, iron support structures shall be earthed by two separate and distinct connections with earth. All the motor couplings exposed rotating parts are to be provided with guard covers for safety of the workforce. • The electrical systems of the hoists and cranes are to be maintained as per IS 3177, IS 4137 and other prescribed standards. Statutory safety inspection needs to be undertaken for the hoists and cranes as applicable.



	• Cables running on the back side/top of the DBs/panels to be properly arranged. All cables to be properly tagged with cable tag at both the ends as well as at intermittent points. HT cables and their routes to be specifically marked at specific intervals. • The cable support structures should be provided with continuous running earth. Power and control cable should be segregated and clearances of power cable from control and communication cables to be maintained as per IS 1255. • All the motors, compressors, machines, metallic parts etc. not intended as conductors, iron support structures etc. shall be earthed by two separate and distinct connections with earth. Frame of every metallic part not intended as conductors, iron support structures shall be earthed by two separate and distinct connections with earth. • Unused openings in control panels/ distribution boards shall be sealed. All panel/board covers shall be connected to earth using flexible copper bonds. • Cable entry or exit into the control rooms, panels, motor terminals, marshalling/console box to be properly done using cable glands and the unused openings to be properly sealed. IS 1255 needs to be adopted for installation and maintenance of cables. • The earthing connection of individual Panels, DBs to be verified for appropriate connectivity with earth. The continuity with earth pits to be verified. • Working space around the control panels and control areas needs to be ensured as prescribed under CEA safety regulation 39. Storage of materials needs to be avoided for the purpose. • Since storage of oil and inflammable items are undertaken as part of the manufacturing process, adequate fire safety measures need to be as per statutory provisions. Use of fire detection and alarms, fire extinguisher systems as applicable needs to be ensured as per prescribed fire hazard classification. Please refer to IS 3594 section 11 regarding electrical installations at storage areas, IS 1641 regarding classification etc. • Since conveyors and other metal fines are being moved in the system, adequate protection needs to be taken against static electricity building up in the conveyors, pipes, structures etc. by providing proper earthing arrangements as per IS 7689. A prescribed maximum resistance of 10 ohm is acceptable to avoid static electricity hazards as per section 15.3 of the IS. Transportable metal items should have a resistance to earth of 10 ohm or less, provided by a special earth connection. • Proper functioning of interlocking and protection arrangements of transformers and other equipment need to be ensured. Appropriate danger notice needs to be provided clearly indicating the voltage level. • All the auxiliary Transformers are to be maintained as per standard and checked for oil leakage. The metal barricades are to be connected to earthing. The metal barricades/enclosures are to be connected to equipotential bonding/earthing. The winding and oil temperature indicators are to be preferably relocated to the front side for safe access, visibility and reading. • Since many outgoing cables are connected to the transformer LV side in some transformers it is advised to check the connections and use suitable cable box or terminal covers. • Cable entry or exit into the control rooms, panels, motor terminals, marshalling/console box to be properly done using cable glands and the unused openings to be properly sealed. IS 1255 needs to be adopted for installation and maintenance of cables. • All cables to be properly tagged with cable tag at both the ends as well as at intermittent points. HT cables and their routes to be specifically marked at specific intervals. • All HV/MV cable inlets/outlets to be properly sealed. All cables to be tagged and transformers to be properly labeled on the HV and MV side with voltage level.
M	DG Sets
	• The DG sets of the 13 MW CPP and Ferro Alloys Plant are to be maintained properly. Power Cables are found to be laid unsafely inside the enclosure. The enclosures are to be cleaned and the damaged acoustic panels are to be repaired. • The DG sets earthing are to be ensured at minimum two points using an exclusive independent earth electrode i.e for the prime mover body and for the generator body and the neutral. Please refer to IS:3043 for further details. The present earthing to be properly verified and measured. • Fire extinguishers and adequate ventilation arrangement is to be made for the DG control room/area. The fire fighting system functioning is to be ensured. • Oil and lubricant leakage if any from the DG sets are to be checked. • The floor of the DG set within the DG enclosure is to be kept clean and dry of spilled oil and lubricants. • It should be ensured that the exhaust system operated with minimum back pressure.



	• The HT cables for the 6 MW DG are to be tagged and identified. Fire retardant paints can be used for safety. Voltage level is to be indicated on the DG Set. Adequate ventilation arrangement is to be made for the DG area. • The asbestos ropes used for the exhaust piping inside the acoustic enclosure/Genset to be periodically checked and replaced to avoid heat input inside the enclosure. • In case the exhaust gas enters the nearby working areas, the direction can be altered or the height can be changed. • The exhaust stack height needs to be verified as per prescribed standard to ensure disposal of exhaust above building height. • All moving parts of the DG set need to be mechanically guarded in such a manner that a human finger cannot reach any moving part. • The space heaters of the DG sets with capacity of more than 500 KVA need to be verified for their operation so that the winding temperature is maintained to avoid moisture during long idle periods. • The interlocking and protection arrangements are to be verified for all the DG sets operating in islanding mode for necessary compliance to regulation 47. • Attention is also invited to the provisions contained in IS 732:2019, specifically the sections 5.5.2.6 and 5.5.2.7, i.e. additional requirements for installations with generating sets providing backup supply or which may operate in parallel with other sources.
N	Floating & Roof Top Solar Plant • <i>A stable, floating approach path for access to the solar panel area needs to be constructed for maintenance and repair purposes in the floating solar plant. Additionally, vegetation growth near the installation area is to be cleaned.</i> • Module serial numbers can be provided for each string. Solar panels are to be cleaned at regular interval; for improved efficiency. Voltage level on inverter panel shall be indicated. • Disconnection switches or circuit breakers provided in combiner boxes to disconnect the PV system from all other conductors of the system shall be located at a readily accessible location and shall be manually operable. • Three phases on the AC side, and positive and negative conductors on the DC side shall be marked and identified with different colors. • The Solar Power Plant shall be equipped with fault ride through capability as per Central Electricity Authority (Technical Standards for Connectivity to the Grid) (Amendment) Regulations, 2019. • Persons designated under regulation 3(1) or engaged or appointed under sub regulation (1) of Regulation 7 or sub regulation (1) regulation 8 shall be provided with dark glasses in addition to the PPEs required as per regulation-21 before entering the solar installations. • Earthing for AC and DC systems shall be interconnected so as to have equipotential bonding. It shall be conforming to IEC 62548. • Lightning protection Installed near the PV modules should be as per approved Indian standard since ESE devices are not recommended by Indian Standard. • Entry to be restricted at every Inverter ITB module. • Proper Segregation and clearance to be provided between positive and negative components including wiring, terminal boxes, etc. inside the combiner box. • All interior and exterior DC conduit, enclosures, cable assemblies, junction boxes, combiner boxes, and disconnectors shall be identified with permanent marking. • All connections shall be properly made through suitable lug/terminal crimped with use of suitable proper cable glands. All cable shall be laid in appropriate cable tray with suitable clearance from ground/roof/wall. • Protective earthing shall be provided inside the inverter cabinet. • All metal casing/shielding of the plant, each array structure of the PV yard, equipment, inverters & control systems shall be grounded through proper earthing. • Every combiner box shall be provided with suitable Surge Protective Device with arc extinguishing capability as per IS 16463 series to avoid any risk of fire. • The input circuits of the combiner box shall be provided with over current protection as per IS/IEC60269-1. • The output circuits of the combiner box shall be provided with isolation protection. • Earth fault protection/ insulation monitoring for PV arrays and inverters shall be provided. • Fire Fighting system for inverter room and control room shall be as per sub-regulation (2) of Regulation 46. • Enclosure of combiners box shall be made of fire retardant material with self-extinguishing property and free from Halogen.



	• Portable fire extinguishers shall also be kept at various locations in plants to take immediate action in case of fire caused by electrical short circuits. • Earthing System shall connect all non –current carrying metal receptacles, electrical boxes, appliance frames, chassis and PV module mounting structures in one long run. The earth strips shall not be bolted. Earthing GI strips shall be interconnected by proper welding. • There shall be adequate numbers (Minimum four (04) numbers) of interconnected earth pit provided in each location. Minimum required gap shall be provided in between earth pits as per relevant standard. • All earth flats used for Lightning Arrestors and solar panel earthing should be embedded to avoid access of human beings. • All earth flats shouldn't come in contact with water pipelines. Necessary precautions to be adapted in this regard. • Three phases on the alternating current side, and positive and negative conductor on the direct current side shall be marked and identified with different colours as per the CEA (Safety) Reg.- 121[1(vi)], 2023. • Shed constructed for inverters & ACDBs shall be earthed from two different earth sources.
O	Building Installations
	• Compliance to the provisions of National Electrical Code, specifically the provisions for lightning protection, earthing, protection against voltage surges, energy efficiency, electrical installation of office buildings, outdoor lighting installations, electrical installation in hazardous areas etc. needs to be considered. • Since most of the buildings have false ceilings, provision of inspection windows/openings needs to be considered to facilitate inspection of electrical connections running within the false ceilings. • Periodical checking of the air conditioning system needs to be ensured, mostly for the stand-alone type air-conditioners so that overloaded operation can be verified to avoid insulation damage due to overheating leading to short circuits. • For proper operation of the circuit breakers/MCCBs, the fault loop impedance of the circuits needs to be verified to ensure proper tripping of the RCDs/MCCBs. • Periodical testing of the lifts, specifically the operation of the ERD needs to be ensured. • Attention is also invited to the provisions of the National Building code, regarding the electrical and allied installations under section 2 Part-8, Volume -I of the Code besides other provisions regarding escalators, lifts, fire and life safety etc. • It is strongly recommended to undertake the energy efficiency study of the buildings to determine the building SEC and take necessary steps to reduce its carbon footprint. • Studies undertaken if any may please be appraised to this office.
13	NB:
	• The installation, operation and maintenance manual of the equipment manufactures is to be strictly adhered to for safe operation. • Considering that this is a Manufacturing plant with furnaces, power plants, SMS etc. adequate safety measures need to be taken for ensuring safety of the workforce. Fire alarms and detection systems need to be provided. Adequate number of fire extinguishers, gas masks and other safety devices at accessible locations needs to be provided. Exit routes to be identified and prominently displayed for use during emergencies. • IS 8062-2 (2006): Code of Practice for Cathodic Protection of Steel Structures, Part II: Underground Pipelines needs to be followed for the safety requirements. • All the motors, compressors, machines, metallic parts etc. not intended as conductors, iron support structures etc. shall be earthed by two separate and distinct connections with earth. Frame of every metallic part not intended as conductors, iron support structures shall be earthed by two separate and distinct connections with earth. All the motor couplings exposed rotating parts are to be provided with guard covers for safety of the workforce. • The Regulation regarding safety procedures, inspection and patrolling, maintenance schedules, use of diagnostic techniques for condition monitoring of equipment, thermo vision scanning, failure analysis, training etc. as prescribed in the CEA (Grid Standards) Regulations 2010 needs to be adopted. • The regulation 48(7) regarding responsibility of the owner needs to be adequately addressed. Failure reporting of transformers as per 48(8) and also the furnishing of statistics as per CEA(Furnishing of Statistics, Returns and Information Regulation, 2007 needs to be followed besides other applicable Regulations and Standards. • Also compliance to the CEA (Technical Standards for connectivity to the Grid) Regulations 2007, CEA (Grid Standard) Regulation 2010 needs to be ensured both as a bulk consumer and also as captive generating stations.



- Please refer to the IS 5613 Part -2, CEA Safety and other Regulations referred herewith besides applicable standards regarding maintenance of the EHT feeders.
- The organization is also advised to furnish periodical returns in prescribed format as per the CEA (Furnishing of statistics, returns and information) regulations 2007.
- Insulating mats are to be installed and maintained as prescribed in the “Indian Standard 15652 (2006): Insulating mats for electrical purposes”, needs to be referred for the purpose.
- Fire extinguishers and detection systems are to be maintained in the control room and other locations as per prescribed Indian Standards i.e. IS 3034, IS 1646, IS 1647, IS 2189, Tariff Advisory Committee manual as applicable.
- The Regulations 29 and 30 of the CEA Safety Regulations 2023 regarding the provision of protective equipment and display of instructions for resuscitation of persons suffering from electric shock is to be strictly followed besides other applicable provisions.
- All equipment, panels shall be marked with the description of its identification at front and at the rear.
- An updated drawing showing the main earth connections and earth electrodes for the electrical installations incorporating the earthing arrangement is to be prepared and maintained. IS 2026, IS 10028, CBIP manual publication no 295 for transformers and other applicable standards needs to be adopted during installation, operation and maintenance of the transformers.
- IS 10118-4 (1982): Code of practice for selection, installation and maintenance of switchgear and control gear, Part 3: Installation, may please be referred for maintenance of switchgears besides other applicable standards such as IS/IEC 62271- 203 (2003): High-Voltage Switchgear and Control gear, Part 203: Gas-Insulated Metal- Enclosed Switchgear for Rated Voltages Above 52 kv [ETD 8: High Voltage Switchgear and Control gear] etc.
- Regular testing of earth value, insulation resistance/IR values, operation of protecting and isolating devices, interlocks etc. to be taken up as per applicable standards to ensure safety and a record of such periodic testing and maintenance is to be recorded in a register and is to be produced during inspection as necessary.
- Safety arrangements to be ensured as per Central Electricity Authority (Safety requirements for construction, operation and maintenance of electrical plants and electric lines) Regulations 2022, CEA (Measures relating to Safety and Electric Supply) Regulations 2023 etc. Specific attention is invited to the regulations 3 to 6 regarding designating and engagement of trained and qualified persons, role of electrical safety officer etc. You may refer to appropriate Regulations of Central Electricity Authority by visiting the CEA website and ensure compliance.
- Reference is also invited to the Central Electricity Authority (Safety requirements for construction, operation and maintenance of electric plants and electric lines) regulations 2022, indicating safety provisions relating to owner, safety manual, safety officer and safety committee, safety provision relating to contractors, reporting of accidents, emergency management plan, medical facilities, safety training and awareness etc. for necessary implementation.
- Required statutory clearances as necessary for operation of the plant needs to be obtained as applicable.
- Deficiencies if any in the report may please be brought to the notice of the undersigned for further necessary action.
- The report in no way relieves the installation owners from any of their statutory obligations and responsibility for ensuring operation and maintenance of the installations in a condition free from danger and as recommended by the manufacturer or by relevant codes of practice of the BIS and CEA.
- The annual inspection of the electrical installations is due from 1st April onward of every year.

It is advised to prepare a SLD exclusively of the metering arrangement of all the TGs/DGs installed along with the metering arrangement including meters installed for recording purchase of power from TPNODL and open access, etc. for review and verification of the dutiable monthly ED. Needs to be submitted on priority.

The metering system of the TGs and DG sets need to be annually tested by STL including the energy meters, associated CTs and PTs. The last test and sealing details are to be submitted to this office for verification and records. Sealing systems are to be maintained properly to avoid any damage to the seals.

M/s JSL is advised to segregate and create separate consumers for the generating units located at JFL and JCL premises. Accordingly the monthly ED dues need to be deposited against the respective consumers.

All the installations covered under this report are considered to be coming under the consumer M/s Jindal Stainless Limited until and unless distinct separation of different entities are legally made as per procedure of statutory provision. This inspection report has been issued considering the electrical safety of electrical installations only inside the existing JSL premises.



Many of the transformers of capacity 10 MVA and higher or with oil capacity of more than 2000 ltrs are found without installation of appropriate fire safety measures and may not be safe to be operated without additional care to avoid fire hazard as mentioned before. The prescribed fire safety measures are to be installed on priority and till such time proactive monitoring of the transformer temperatures and loads are to be ensured and in case of any abnormal temperature rise, the transformers are to be shut down immediately. The support of manufacturers in this regard may be obtained. Also, oil leakages in many transformers are still observed which increases the fire hazard and need to be addressed on priority.

The electrical installations covered in this report are allowed to remain energized and compliance and rectification of defects and deficiencies recommended in the report needs to be submitted to this office within 90 days of issue of this report.

Digitally Signed

Er. A. Behera

C.E-Cum-C.E.I., North Zone
Odisha, Balasore

ANNEXURE-A

TG/DG Set Energy Meter Readings (07.03.2026)

TG/DG	TG1	TG2	TG3
Make:-	BHEL/ 1536	BHEL/ 1538	BHEL
VOLTAGE	11kV 3 Phase	11kV 3 Phase	6.6 KV 3 Phase
Capacity	125 MW	125 MW	13 MW
Energy Meter make and sl no	L&T, 06606965	L&T, 06606975	SECURE, ABB00283
Energy meter reading	14408948 MWH	14585602 MWH	421052 MWH

The energy metering system (Energy Meter, CTs, PTs) of the TG1 and TG2 are to be replaced with 0.2s class energy metering system on priority.

Sl. No.	Location	DG/TG	Capacity	Meter Reading
1	2*125MW	DG-1	600 KVA	4.0 KWH
2	2*125MW	DG-2	600 KVA	2.6 KWH
3	Stainless House	DG-3	600 KVA	614.4 KWH
4	Fe-Cr	DG-4	600 KVA	32.7 KWH
5	Fe-Mn	DG-5	600 KVA	36.1 KWH
6	Manpur	DG-6	600 KVA	2975.4 KWH
7	13MW	DG	750 KVA-1	4.5 KWH
8	13MW	DG	750 KVA-2	4.8 KWH
9	CRM-CAPL	DG	1010 KVA-1	20.52 KWH
10	CRM-HAPL	DG	1010 KVA -2	14.44 KWH
11	MRSS	DG	82.5 KVA	DG NOT IN RUNNING CONDITION
12	SMS PH	DG	1500 KVA-1	10.59 MWH
13	SMS S/S	DG	1500 KVA-2	15.75 MWH.
14	HSM	DG	1500 KVA-3	9.21 KWH
15	COBP	DG	6 MW	DG NOT IN RUNNING CONDITION
16	COBP	DG	125 KVA	1215 KWH
17	COMBO LINE	DG	1010 KVA	4.06 MWH
18	PELLET PLANT	DG	625 KVA	3.23 MWH
19	FLOATING SOLAR	SOLAR	7.32 MWp	20215.8 MWH

It is observed that monthly ED returns for some of the newly installed DG sets are not being communicated. It needs to be communicated. by JSL. Energy Meter readings of the rooftop solar plant meters could not be taken. M/s JSL is advised to furnish the details along with current seal details for further verification and record.

The meter, CT, PT, the testing and sealing details are to be submitted for all DG sets for necessary verification and updating of records. Defective energy meter noticed if any are to be intimated to this office and to be replaced on priority after necessary approval of this office.



Discrepancy if any may please be brought to the notice for necessary action. Integrity of sealing arrangements for the TG/DG Meters and CTs/PTs are to be ensured and the seals are to be prevented from damage. The current seal details of all TG/DG sets need to be furnished for verification and record. Energy meters along with CT needs to be tested annually by STL and test reports need to be submitted to this office for records.

The energy metering system (Energy Meter, CTs, PTs) of the TG1 and TG2 are to be replaced with 0.2s class DLMS compliant energy metering system on priority.

ANNEXURE-B

PROVISIONAL ANNUAL INSPECTION FEES FOR THE FY: 2025-26 OF M/S. JINDAL STAINLESS LTD, JAJPUR INELEIN/2026/00333						
SI No	Equipment Description	Rating	Unit	Qty	Rate in Rs.	Amount in Rs.
I. 220 KV MRSS						
220 KV Switchyard						
1	Transformer, 120 MVA, 220/34.5kV	120	Nos	3	260000	780000
2	Transformer, 63/80 MVA, 220/34.5kV	80	Nos	5	215000	1075000
3	Additional Bays	220 KV	Nos	8	20000	160000
4	220 kV Conductor (12 KM)	220 KV	KM	12	6500	6500
33 KV Switchgear at MRSS						0
1	Transformer, 0.5 MVA, 33/0.433kV	0.5	Nos	2	7000	14000
2	33 KV Auxiliary Board (Bay)	33 KV	Nos	20	20000	400000
3	33 kV Mill Board (Bay)	33 KV	Nos	18	20000	360000
4	Power Cable 33 KV (13 KM)	33 KV	KM	13	14000	14000
5	Control Cable	415 V	KM	67	8000	8000
6	Miscellaneous Load		KW	120	13000	13000
33 KV SS-2						0
1	Transformer, 10/12 MVA, 33/6.6kV	12	Nos	1	50000	50000
2	Transformer, 6.6/0.43 KV, 1.6MVA	1.6	Nos	1	15000	15000
3	Additional Switchgears 6.6 kV	6.6 KV	Nos	9	1000	9000
4	Power cable 33 kV (8 KM)	33 KV	KM	8	9000	9000
5	Control Cable	415 V	KM	1	250	250
Sub-Station-3(Near Metal Recovery Plant)						0
1	Transformer, 33/6.6 KV, 10/12 MVA	12	Nos	1	50000	50000
2	Transformer, 6.6/0.43 KV, 1.6MVA	1.6	Nos	1	15000	15000
3	6.6 KV switchgear panel	6.6 KV	Nos	4	1000	4000
4	33 KV Extension Panel Bay with IS Limiter	33 KV	Nos	3	20000	60000
II. COBP						0
1	Transformer, 10/12 MVA, 33/6.6kV	12	Nos	2	50000	100000
2	Transformer, 6.6/0.433 KV, 0.5 MVA	0.5	Nos	1	7000	7000
3	Transformer, 6.6/0.690 KV, 0.8 MVA	0.8	Nos	1	7000	7000
4	Transformer, 6.6/0.433 KV, 1.6MVA	1.6	Nos	2	15000	30000
5	Transformer, 6.6/0.433 KV, 1 MVA	1	Nos	10	7000	70000
6	Capacitor 2500 KVAR, 6.6 KV		No	1	2500	2500
7	Additional Switchgears 6.6 kV & I/c Panel		Nos	12	1000	12000
8	Power cable 33 kv (24 KM)		KM	24	25000	25000
9	LT Cable		KM	197	17000	17000
10	Aggregate Load		MW	17.5	87500	87500
11	8 MVA, 11/0.4 KV Transformer (DG Station)	8	Nos	1	32500	32500
12	1 MVA, 11/0.4 KV Transformer (DG Station)	1	Nos	1	7000	7000
13	LT Switchgear (PCC/VFD/MCC/PDB) Panel		Nos	76	500	38000
III. CRM						0
1	2 MVA 33/.440 kv transformer	2	Nos	24	15000	360000
2	4.2 MVA 33/.440 kv transformer	4.2	Nos	1	15000	15000
3	1.6 MVA 33/ .440 kv transformer	1.6	Nos	6	15000	90000
4	5.8 MVA 33/3.3 KV transformer	5.8	Nos	2	32500	65000
5	33 KV HT Bay (BC, PT, BR)		Nos	12	20000	240000
6	Power Cable 33 KV (6 KM)			1	7000	7000
7	VDF 3.3 KV		Nos	10	1000	10000
8	LT Cable		KM	90	10000	10000
9	Aggregate Load		MW	60.05	125000	125000



10	LT Switchgear (PMCC/PCC/MCC/VFD/PDB) Panel		Nos	532	500	266000
IV. CRM (NEW)						0
1	7.5 MVA, 33/3.3 KV Converter Tr.	2	Nos	2	32500	65000
2	2 MVA, 33/0.415 KV Distribution Tr.	2	Nos	4	15000	60000
3	2 MVA, 33/0.433 KV Dry Type Dist/ Tr.	2	Nos	2	15000	30000
4	4.2MVA 33/6.6KV Transformer	4.2	Nos	1	15000	15000
5	382 KVA, 415/220 V Excitation Tr.	0.382	Nos	2	7000	14000
6	33 KV HT Bay		Nos	7	20000	140000
7	6.6 KV Panels		Nos	1	1000	1000
8	PCC/MCC/LDB/PLC/RIO/UPS/SERVER Panels		Nos	22	500	11000
9	HT Cable 2 KM			1	3000	3000
10	LT Control/Power Cable		KM	586	37000	37000
11	Aggregate Load		MW	33.26	102500	102500
V. CRM COMBO LINE						
1	5.8MVA, 33/3.16 KV Converter Transformer	5.8	Nos	2	32500	65000
2	2.5MVA, 33/0.415KV Distribution Transformer	2.5	Nos	1	15000	15000
3	2MVA, 33/0.415KV Converter Transformer	2	Nos	9	15000	135000
4	2MVA, 33/0.415KV Distribution Transformer	2	Nos	8	15000	120000
5	33 KV Entry/Exit Bay	33 KV	Nos	28	20000	560000
6	3.3 KV Drives Panel		Nos	1	1000	1000
7	MV,LT Drive/PMCC/PCC/MCC/EMCC/PDB Panel		Nos	48	500	24000
8	HT cable 300 SQ MM XLPE, 33 KV		KM	6	7000	7000
9	LT Cable		KM	350	27000	27000
10	Aggregate Load		MW	36.031	102500	102500
VI. CRM FINISHING LINE						
1	2.5MVA, 33/0.720KV Converter Transformer	2.5	Nos	1	15000	15000
2	2 MVA, 33/0.415KV Distribution Transformer	2	Nos	2	15000	30000
3	HT cable 300 SQ MM XLPE, 33 KV		KM	2	3000	3000
4	HT/LT Cable		KM	35	5000	5000
5	Panels/VFD		Nos	111	500	55500
6	Aggregate Load		MW	8.327	73000	73000
VII. HSM						0
1	9 MVA 33kv/3.3kv FM 1 to FM # 7	9	Nos	7	32500	227500
2	9 MVA 33kv/3.3kv RM TOP AND RM BOTTOM	9	Nos	2	32500	65000
3	4 MVA 33kv/3.3kv	4	Nos	1	15000	15000
4	2 MVA 6.6kv/415v	2	Nos	16	15000	240000
5	2.5 MVA 6.6kv/415v	2.5	Nos	12	15000	180000
6	31.5 MVA 33kv/6.6kv	31.5	Nos	2	115000	230000
7	10/12.5 MVA 33kv/6.6kv	12.5	Nos	4	50000	200000
8	1.6 MVA 6.6kv/415v	1.6	Nos	4	15000	60000
9	0.500 MVA 6.6kv/415v Lighting Tr.	0.5	Nos	3	7000	21000
10	0.315 MVA 415/415v Lighting Transformer	0.315	Nos	1	7000	7000
11	25 KVA, 415/230v	0.025	Nos	11	5000	55000
12	HT 33 KV Bay (OG, PT, BC)		Nos	8	20000	160000
13	6.6 KV Switchboard Panels		Nos	20	1000	20000
14	3.3 KV Drives Panel		Nos	9	1000	9000
15	MV VFD		Nos	9	500	4500
16	LT Switchgear panels		Nos	625	500	312500
17	Power cable 33 kv (3 km)			1	4000	4000
18	Power cable up to 11 kv & Control Cable (1099 KM)			1	62000	62000
19	Aggregate Load		MW	134.75	170000	170000
Down Coiler-2, WCS Expansion						
1	2.5 MVA DC-2 Main Drive Transformer	2.5	Nos	1	15000	15000
2	2 MVA DC-2 PCC transformer	2	Nos	1	15000	15000
3	Aggregate Load		MW	7.16	68000	68000
4	LT Switchgear panels		Nos	26	500	13000
5	11KV/6.6KV/LT Control & Power Cable		KM	46.23	6000	6000
VIII. GENERATION PLANT						0
1	125 MW Turbo Generator-1 & 2	125	Nos	2	925000	1850000
2	13 MW Turbo Generator	13	Nos	1	125000	125000
3	600 KVA DG Set (CPP-1, CPP-2, Stainless House, Fe-Cr, Fe-Mn)	0.6	Nos	5	7000	35000
4	750 KVA DG Set (13 MW TG)	0.75	Nos	2	7000	14000
5	1010 KVA DG Set (CRM-CAPL, HAPL, Comboline)	1.010	Nos	3	7000	21000



6	1500 KVA DG Set (HSM PH, SMS S/S, HSM)	1.5	Nos	3	25000	75000
7	6 MW COBP DG-1	6	No	1	75000	75000
8	125 KVA COBP DG-2	0.125	Nos	1	5000	5000
9	82.5 KVA DG Set	0.0825	Nos	1	5000	5000
10	652 KVA Chrome Pellet Plant DG	0.625	No	1	7000	7000
11	7.32 MWp Floating Solar Power Plant	7.32	No	1	75000	75000
12	14.4 MWp Rooftop Solar Power Plant (13 Nos of location)	14.4	No.	1	220000	220000
13	8.62 MWp HSM Rooftop Solar Plant (3 Nos. of Locations)	8.62	No.	1	169500	169500
IX. FE-ALLOYS						0
1	33/6.6 KV 10/12 MVA Transformer	12	Nos	3	50000	150000
2	Transformer, 2 MVA, 6.6/0.415kV	2	Nos	6	15000	90000
3	Transformer,1.6 MVA, 6.6/0.415kV	1.6	Nos	6	15000	90000
4	Transformer,9.2 MVA,SAF 3,4&5	9.2	Nos	9	32500	292500
5	Transformer,20 MVA,SAF 1&2	20	Nos	6	67500	405000
6	Transformer,0.5 MVA, 6.6/0.415kV	0.5	Nos	1	7000	7000
7	CAPACITOR BANK 20 MVAR,33KV	20	Nos	3	2500	7500
8	CAPACITOR BANK 10 MVAR,33KV	10	Nos	3	2500	7500
9	CAPACITOR BANK 2.5 MVAR,6.6 KV	2.5	Nos	2	2500	5000
10	CAPACITOR BANK 12.96 MVAR	12.96	Nos	1	2500	2500
11	CAPACITOR BANK 9.72MVAR	9.72	Nos	2	2500	5000
12	CAPACITOR BANK 6.48 MVAR	6.48	Nos	3	2500	7500
13	HT 33 KV Bay (BC, BPT, BR)		Nos	25	20000	500000
14	Power cable 33KV (12 KM)	33 KV		1	13000	13000
15	Cable up to 11 KV (291 KM)			1	22000	22000
16	0.433 KV Switchgear Panels		Nos	55	500	27500
17	Aggregate Load		MW	8.87	73000	73000
ETP						0
1	Cable up to 11 KV (48 KM)			1	6000	6000
2	MCC/PD/PLC Panel		Nos	5	500	2500
3	Aggregate Load		MW	0.875	30500	30500
SAF 4&5 New Dedusting System & Fume Extraction System						
1	Cable up to 11 KV (67 KM)			1	8000	8000
2	MCC/PD/PLC Panel		Nos	10	500	5000
3	Aggregate Load		MW	1.071	38000	38000
X. Chrome Pellet Plant						
1	2.5 MVA 6.6/433 KV TRANSFORMER	2.5	Nos	2	15000	30000
2	76 KVA ESP TRANSFORMER RECTIFIER	0.076	Nos	4	5000	20000
3	HT 6.6 KV Panel / Rotor Control/ VFD		Nos	16	1000	16000
4	LT Switchgear Panel		Nos	83	500	41500
5	6.6 KV/LT Cable		KM	53.7	7000	7000
6	Aggregate Load		MW	7.11	68000	68000
XI. SMS						0
1	120 MVA, Furnace Transformer, 33 KV/595 V	120	Nos	1	260000	260000
2	100 MVA, Furnace Transformer, 33 kv/1000 V	100	Nos	1	265000	265000
3	24 MVA, Lrf Transformer, 33 KV/380 V	24	Nos	1	85000	85000
4	16 MVA Distribution trafo, 33 KV /6.6 KV	16	Nos	5	67500	337500
5	4 MVA Distribution trafo 6.6 KV/0.690 KV	4	Nos	3	15000	45000
6	2 MVA Distribution trafo. 6.6 KV/0.433 KV	2	Nos	8	15000	120000
7	1.6 MVA Distribution trafo. 6.6 KV/0.433 KV	1.6	Nos	9	15000	135000
8	2 MVA Dry Type Distribution trafo. 6.6 KV/0.433KV	2	Nos	7	15000	105000
9	3 MVA, Induction Furnace Transformer,6.6/0.85KV	3	Nos	1	15000	15000
10	1 MVA Dry type Distribution trafo,6.6 KV/0.433 KV	1	Nos	1	7000	7000
11	315 KVA DRY Special 6.6KV/0.433 KV	0.315	Nos	1	7000	7000
12	33 KV,45 MVAR Capacitor Bank	45	Nos	2	13000	26000
13	33 KV,40 MVAR Capacitor Bank	40	Nos	1	13000	13000
14	33 KV,30 MVAR Capacitor Bank	30	Nos	1	13000	13000
15	Capacitor Unit(300 KVAR),415V	0.3	Nos	1	13000	13000
16	HT 33 KV Bay (BC, PT, TIE, CICO)		Nos	10	20000	200000
17	HT 6.6 KV Panel / Rotor Control/ VVF		Nos	24	1000	24000
18	LT PCC/MCC/PDB/VVF/RIO Panel		Nos	221	500	110500
19	Aggregate Load		MW	242.91	190000	190000
EQUIPMENT LIST OF 6T INDUCTION FURNACE FOR (SMS)						0
1	HT/LT Cable (91 KM)			1	11000	11000



2	LT PCC/MCC Panel		Nos	3	500	1500
3	HT cable (33 KV)(4Km)	33 KV		1	5000	5000
BOC PLANT (OXYGEN PLANT)						0
1	15 MVA, 33/6.6 KV Transformer	15	Nos	1	50000	50000
2	2.5 MVA, Transformer 33/0.433 KV	2.5	Nos	1	15000	15000
3	Capacitor 1.4MVAR bank	1.4	Nos	2	2500	5000
4	HT 33 KV Bay (BC, BPT, BR)		Nos	3	20000	60000
5	HT 6.6 KV Panel		Nos	7	1000	7000
6	LT PCC/MCC Panel		Nos	24	500	12000
7	Power Cable 33 kv (1 km)			1	2000	2000
8	LT Cable		KM	6	1000	1000
9	Aggregate Load		MW	11.83	87500	87500
METAL RECOVERY PLANT						0
1	Transformer 2 MVA (6.6 KV/415 V)	2	Nos	1	15000	15000
2	HT Capacitor bank,6.6 KV,172 KVAR	0.172	Nos	2	1000	2000
3	Capacitor bank,0.415 KV,50 KVAR	0.050	Nos	2	1000	2000
4	Capacitor bank,0.415 KV,25 KVAR	0.025	Nos	2	1000	2000
5	HT/LRS Panel		Nos	3	1000	3000
6	LT PCC/MCC Panel		Nos	33	500	16500
7	HT/LT cable upto 11 KV		KM	34	5000	5000
8	Aggregate Load		MW	2.632	43000	43000
SMS (Additional)						
1	LRF -2 Transformer, 33 KV/0.385 KV	24	Nos	1	85000	85000
2	2 MVA Dry Type Distribution trafo. 6.6 KV/0.433KV @ Caster-2 & AOD2	2	Nos	3	15000	45000
3	1MVA DRY Special 6.6KV/0.710KV @ Caster-2	1	Nos	1	7000	7000
4	0.8MVA Distribution trafo. 6.6 KV/0.433 KV @BOC	0.8	Nos	1	7000	7000
5	6.6 KV Panels		Nos	4	1000	4000
6	LT SWITCHGEAR Panels		Nos	43	500	21500
7	LRF -2 33KV cable For Transformer(1600mtrs)			1	2000	2000
8	CCM2,AOD2,LRF2,BOC (TOTAL Length=4500mtrs)			1	850	850
9	Aggregate Load		MW	29.19	167500	167500
XII. CENTRAL UTILITY						0
1	1.6 MVA, Transformer -1	1.6	Nos	1	15000	15000
2	2 MVA, Transformer -2	2	Nos	1	15000	15000
3	LT SWITCHGEAR Panels		Nos	13	500	6500
3	HT/LT cable upto 11 KV		KM	5.7	1000	1000
4	Aggregate Load		MW	2.705	43000	43000
XIII. CRMHS						0
1	Transformer, 10/12 MVA, 33/6.6kV	12	Nos	1	50000	50000
2	Transformer,6.6/0.433 KV,2 MVA	2	Nos	6	15000	90000
3	Transformer,6.6/0.433 KV,0.5MVA	0.5	Nos	1	7000	7000
4	Transformer,6.6/0.433 KV,1.6MVA	1.6	Nos	2	15000	30000
5	Transformer,6.6/0.433 KV,1 MVA	1	Nos	2	7000	14000
6	Capacitor 510 KVAR, 6.6 KV	0.51	Nos	4	2500	10000
7	6.6 KV Switchgear Panels		Nos	5	1000	5000
7	Power Cable 33 KV		KM	0.7	2000	2000
8	LT Cable		KM	115	17000	17000
9	Aggregate Load		MW	5.68	58000	58000
A	Annual Inspection Fees for the FY: 2025-26					16216100
NB: - The amount is provisional based on details submitted and may be revised if necessary after field verification and in case additions and alterations are noticed subsequently which are not included in the list.						